

Number Representations

This worksheet will give you some practice converting and working with decimal, binary, and hexadecimal numbers. You can also raise your hand and ask for help if you are stuck!

1. Convert the decimal number 17 to binary and hexadecimal.
2. Convert the hexadecimal number `0xf` to binary and decimal.
3. Convert the binary number `0b1010` to decimal and hexadecimal.
4. Look at your answers to question 2 and 3. How many binary digits is needed to represent a single hexadecimal digit?
5. If a byte is 8 bits, how many hexadecimal digits does it take to represent a byte?
6. **Challenge:** Now that you know how to convert between decimal, binary, and hexadecimal, try applying bitwise operators onto these numbers:
 - `0b1 << 2 =`
 - `8 & 1 =`
 - `0xa | 2 =`

7. Adding and subtracting numbers in different number systems can be a bit tricky at first, but it follows the same general rules as adding and subtracting in base 10.

Alternatively, you can convert back to base 10 and then perform the operation.

Challenge: Answer the following questions

- `0b0010 + 0b1001 =`
- `0xa - 0x5 =`
- `0xf + 1 =`
- `0b1100 - 0xb =`